

Linear Regression and Logistic Regression

Your Name

- Linear Regression intuition
- Closed form solution using cov and var
- Numeric example
- Transition to Logistic Regression

Linear Regression

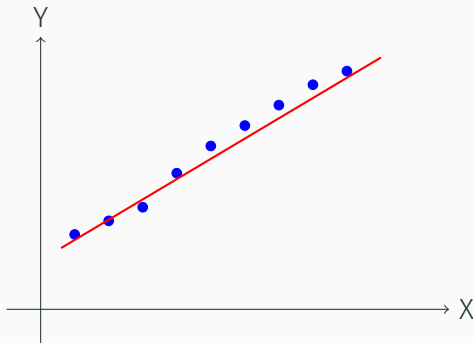
Goal of Linear Regression

- Fit a straight line

$$\hat{y} = wx + b$$

- Minimize squared error

$$J = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$



Closed Form: Using Covariance and Variance

Slope:

$$w = \frac{\text{cov}(x, y)}{\text{var}(x)}$$

Intercept:

$$b = \bar{y} - w\bar{x}$$

Where:

$$\text{cov}(x, y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$\text{var}(x) = \frac{1}{n} \sum (x_i - \bar{x})^2$$

Numeric Example

Given the dataset:

$$(x, y) = \{(1, 2), (2, 3), (3, 5)\}$$

Step 1: Means

$$\bar{x} = 2, \quad \bar{y} = \frac{2 + 3 + 5}{3} = 3.33$$

Step 2: Covariance

$$\begin{aligned} \text{cov}(x, y) &= \frac{(1 - 2)(2 - 3.33) + (2 - 2)(3 - 3.33) + (3 - 2)(5 - 3.33)}{3} \\ &= \frac{(-1)(-1.33) + 0 + (1)(1.67)}{3} = \frac{1.33 + 1.67}{3} = 1 \end{aligned}$$

Step 3: Variance

$$\text{var}(x) = \frac{(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2}{3} = \frac{1 + 0 + 1}{3} = \frac{2}{3}$$

Step 4: Slope

$$w = \frac{1}{2/3} = 1.5$$

Step 5: Intercept

$$b = \bar{y} - w\bar{x} = 3.33 - 1.5(2) = 0.33$$

Final model:

$$\hat{y} = 1.5x + 0.33$$

Why Linear Regression Fails for Classification

Linear regression outputs can be negative or above 1.

Classification requires probability:

$$0 \leq P(y = 1|x) \leq 1$$

We need a constrained output.

Logistic Regression

Logistic Regression Model

Sigmoid:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

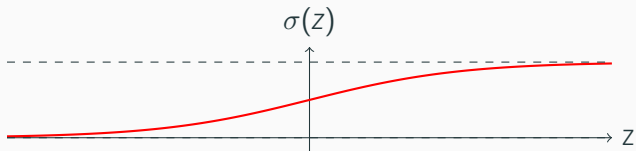
Model:

$$\hat{y} = \sigma(wx + b)$$

Interpretation:

$$\hat{y} = P(y = 1|x)$$

Sigmoid Curve



$$J = -\frac{1}{n} \sum [y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

Optimized using gradient descent.

Comparison

Aspect	Linear	Logistic
Target	Continuous	Binary
Output Range	any real number	0 to 1
Loss	MSE	Log Loss
Interpretation	Predicts value	Predicts probability

Questions?